

IOWA COUNTY AP, WI, USA

WMO: 726507

Lat: 42.887N

Lon: 90.236W

Elev: 1171

StdP: 14.08

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 54907

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.1	-0.3	-14.7	2.6	-4.7	-8.9	3.6	1.0	25.0	24.4	22.6	21.6	7.5	290	0.523

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	89.5	73.6	86.1	72.6	82.5	70.9	78.0	84.3	75.8	82.5	74.0	80.5	10.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.5	139.4	81.8	73.2	129.0	79.6	72.2	124.4	78.7	42.6	85.3	40.1	83.4	38.4	80.4	88.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.5	19.9	17.9	-13.2	93.1	6.2	2.8	-17.6	95.1	-21.2	96.8	-24.7	98.4	-29.2	100.4		
(5)			-13.1	80.5	5.4	2.2	-17.0	82.1	-20.2	83.4	-23.2	84.6	-27.1	86.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.3	20.9	24.2	35.7	48.0	59.4	68.5	72.4	70.3	63.4	50.6	38.4	26.0		
	DBStd	20.33	11.87	11.54	11.57	9.48	8.64	6.52	5.67	5.29	7.97	9.24	9.82	11.45		
	HDD50	3479	903	723	465	146	18	0	0	0	4	107	367	746		
	HDD65	6795	1368	1143	909	514	218	39	7	14	123	453	798	1209		
	CDD50	2847	0	1	22	86	308	554	695	629	405	127	18	2		
	CDD65	689	0	0	1	4	43	143	238	178	74	8	0	0		
	CDH74	5454	0	0	10	52	371	1137	1983	1262	578	61	0	0		
	CDH80	1498	0	0	0	8	86	305	611	326	155	7	0	0		
	WSAvg	8.1	8.8	8.9	9.4	10.2	8.6	6.9	6.0	5.6	6.8	8.1	8.9	8.7		
	PrecAvg	38.1	1.3	1.4	2.2	3.7	4.3	5.8	4.5	4.6	3.5	2.6	2.4	1.6		
(15) Precipitation	PrecMax	48.3	2.5	2.9	5.6	7.2	10.4	10.4	9.4	17.2	7.2	5.6	7.4	4.5		
	PrecMin	28.9	0.2	0.1	0.4	1.4	0.8	1.1	1.8	1.5	0.6	0.6	0.3	0.2		
	PrecStd	5.9	0.6	0.8	1.4	1.7	2.0	2.6	1.9	3.3	1.7	1.5	1.9	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.3	56.9	73.4	80.8	88.2	90.5	94.8	90.6	89.7	80.9	68.4	57.5		
		MCWB	42.4	51.0	61.8	61.2	68.8	74.0	73.6	71.8	71.5	66.7	57.3	55.5		
	2%	DB	43.1	48.1	64.1	73.0	81.7	86.4	89.7	86.4	84.3	74.8	62.6	50.2		
		MCWB	39.1	44.5	55.7	58.3	65.7	71.6	76.1	73.0	70.9	63.5	54.7	46.6		
	5%	DB	37.4	43.3	58.7	68.4	78.7	82.8	86.1	82.5	80.9	70.3	57.2	44.7		
		MCWB	35.8	39.3	51.3	55.6	63.9	70.7	74.0	71.4	68.1	61.3	51.1	41.7		
	10%	DB	36.1	38.6	52.2	63.7	73.2	81.0	82.4	81.1	76.9	65.7	53.7	39.3		
		MCWB	34.6	36.1	45.8	52.5	62.1	69.2	72.4	70.1	65.9	59.3	48.4	36.8		
	0.4%	WB	44.8	53.6	63.0	65.1	72.7	79.6	82.1	78.2	76.3	73.5	60.3	56.6		
		MCDB	46.0	55.2	72.7	75.1	82.8	83.8	86.4	84.4	83.8	78.4	65.3	58.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	39.0	45.2	57.5	61.4	69.5	76.2	79.3	76.0	73.3	67.2	56.1	46.5		
		MCDB	42.5	47.4	62.7	69.0	77.5	83.3	85.2	82.8	79.9	70.7	60.8	48.1		
	5%	WB	36.6	39.5	51.8	58.1	66.9	73.7	76.8	74.2	71.0	62.9	52.8	41.4		
		MCDB	38.3	42.6	56.9	66.5	75.2	80.6	83.2	80.7	78.0	67.5	56.9	44.3		
	10%	WB	34.3	36.5	45.9	53.8	64.2	71.2	74.5	72.3	68.3	58.9	47.7	37.3		
		MCDB	35.7	38.4	50.7	61.5	71.5	78.0	80.9	78.5	74.8	65.0	52.2	39.3		
	5% DB	MDBR	13.5	14.3	15.5	18.4	18.5	17.7	17.6	17.4	19.3	17.3	14.4	12.5		
		MCDBR	14.2	16.9	22.2	25.4	22.3	20.4	19.5	19.8	22.6	22.9	20.5	15.2		
		MCWBR	11.7	12.9	14.0	14.6	11.7	10.8	9.6	9.7	11.1	13.3	14.5	12.8		
		MCWBR	14.0	16.5	20.4	22.4	19.2	18.2	17.1	17.2	19.3	19.7	19.4	15.0		
(40) Clear-Sky Solar Irradiance	taub		0.274	0.300	0.321	0.365	0.392	0.412	0.419	0.404	0.372	0.331	0.292	0.274		
		taud	2.392	2.322	2.409	2.315	2.291	2.270	2.251	2.295	2.366	2.480	2.517	2.439		
	Ebn at Noon		276	285	293	286	280	273	270	270	272	272	269	266		
		Edh at Noon	24	31	33	39	41	42	43	39	34	27	22	22		
(44) All-Sky Solar Radiation	RadAvg		557	884	1204	1452	1717	1917	1930	1681	1383	894	588	431		
	RadStd		68	92	132	121	124	147	111	113	111	111	47	38		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (51 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page